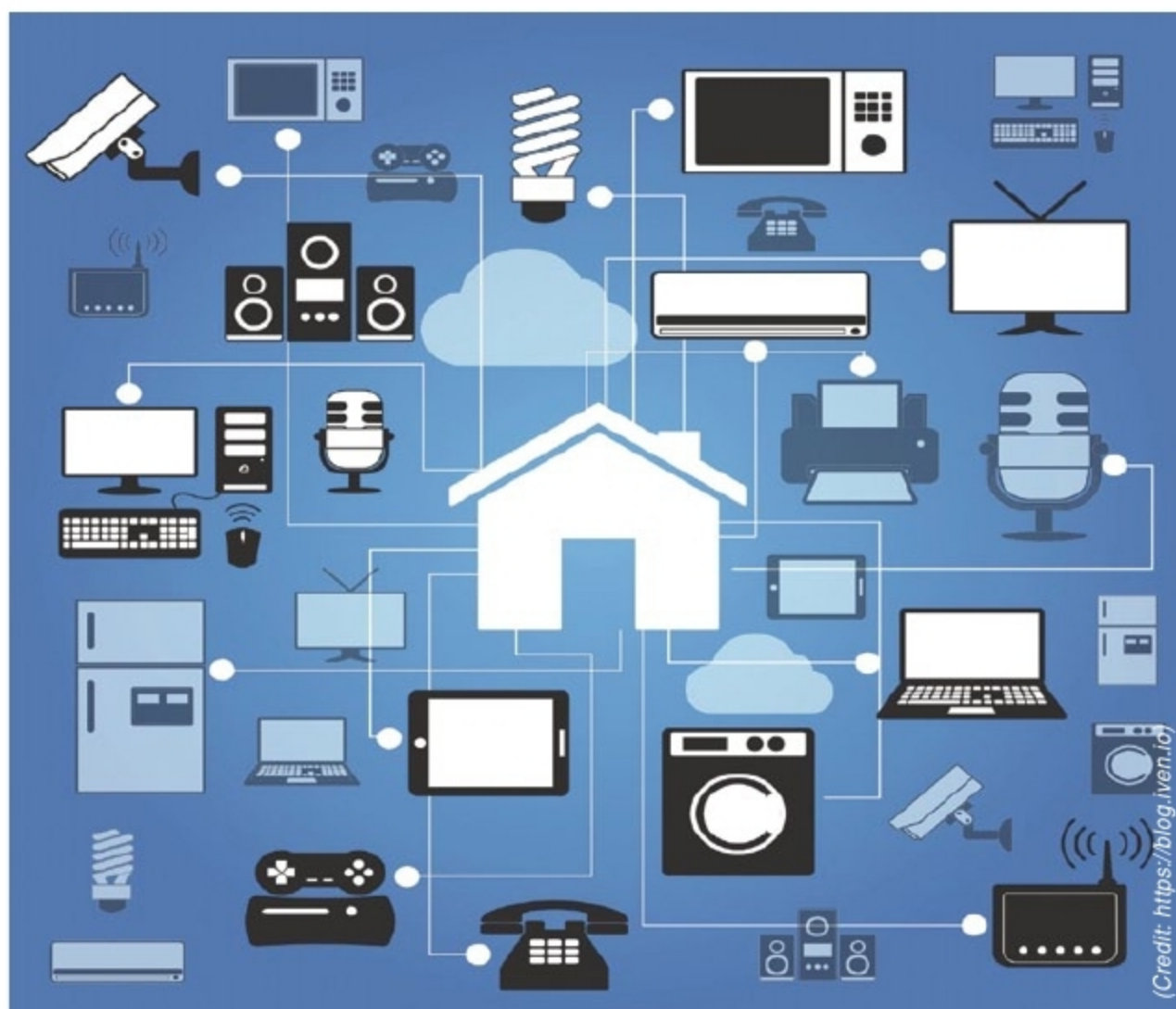


A Guide To The **ESSENTIAL** **PROTOCOLS** For The IoT

The IoT extends from a single constrained device to a whole range of cloud systems, all connected by a set of IoT protocols that allow devices and servers to talk to one another



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When you surf the Internet or purchase a cellphone on, say, Flipkart, you use Hyper Text Transfer Protocol Secure (HTTPS), Transmission Control Protocol (TCP), Domain Name System (DNS), IPv6 or IPv4 data link layer protocols and 802.3 protocols. These are basic networking protocols inside a browser, computer's operating system and asymmetric digital subscriber line (ADSL) router. You use these protocols even while checking an email on your computer.

Protocols are the building blocks of all networking technologies on the Internet. These are software, and size of these protocols in Windows operating system is in megabytes. Researchers all over the world are trying to reduce the size of TCP/IP protocols.

HTTP is a software protocol, which comes inbuilt in the Internet browser, such as Chrome. It sends commands to the Web server and gets replies from the same. This software listens on port number 80 and receives IP packet on 10.100.10.11 IP address.

Let us take an example. Flipkart uses the secure HTTPS protocol to communicate with the browser. This is a new protocol. It listens on port number 443 and 10.100.10.11 IP address. (Port number is a 16-bit logical number between 0 and 65535.) Here, IP address did not change, but port number did.

Port number and IP address form the socket. Hence, the Internet uses socket for communication. A secure socket opens between the browser and Flipkart's Web server. DNS protocol resolves *www.flipkart.com* string to